

Prathmesh Jain

+91 8657376595 | prathmeshjain42@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

EDUCATION

Vidyalankar Institute of Technology

Bachelors in Computer Engineering

GPA: 9.2 / 10

Mumbai, IN

May 2024

K.C. College

12th Science

79.38%

Mumbai, IN

May 2020

EXPERIENCE

Tata Consultancy Services (TCS)

Systems Engineer

Oct 2024 – Present

- **AI-driven Compliance Screening Platform** – Designed and implemented an agentic compliance screening system using FastAPI (SSR), Azure OpenAI, and LangChain integrated with Model Context Protocol (MCP) tools such as Google Search and custom web crawlers.
- Enabled the LLM to autonomously decide which MCP tools to invoke and when, creating a fully agentic workflow for customer verification and adverse media screening.
- Implemented sanctions and embargo verification using fuzzy matching, reducing false positives and manual intervention.
- Optimized data ingestion and processing pipelines using asynchronous I/O and caching, improving overall throughput and responsiveness for large-scale Excel datasets.
- **Credit Memo Generation System** – Developed an AI-assisted financial analysis solution capable of generating credit memos directly from company financial reports.
- Implemented a hybrid design: LangChain-based pipelines for financial spread generation (structured numerical extraction) and a CrewAI-powered agentic layer for autonomous credit memo synthesis.
- Processed and chunked financial data, indexed it with vector embeddings, and utilized LLM-based reasoning to derive contextual insights.
- Orchestrated components using FastAPI, ensuring modularity, secure LLM interactions, and smooth data flow between retrieval and generation stages.

PROJECTS

Patronix – E-commerce Platform | *React.js, Node.js, Express.js, MongoDB*

July 2023

- Developed a full-stack E-commerce web application with user authentication, product management, and order tracking.
- Optimized frontend performance and global state management using Redux Toolkit.
- Designed RESTful APIs with token-based authentication and modular architecture for maintainability.

ExpenseGauge – Expense Tracker App | *React Native, Node.js, MongoDB*

March 2024

- Developed a mobile expense tracking app with offline-first support using Zustand for state management.
- Implemented secure authentication with rotating jwt tokens and background data synchronization.
- Visualized spending patterns with dynamic charts.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS / NativeWind

Backend: Node.js, Express.js, FastAPI, Flask

Database: MongoDB, PostgreSQL

Tools: Docker (Learning), Git, Postman, Playwright